Similarity

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What is similarity?

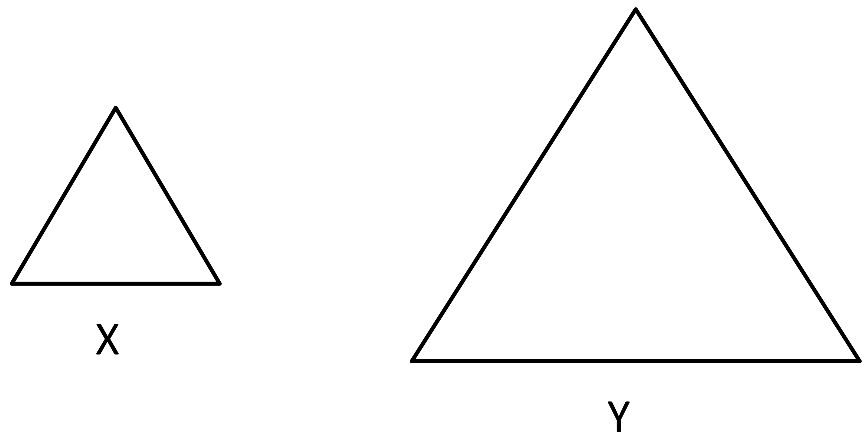
The most fundamental question of **cognition**, I presume, is the measurement of similarity.

Though a simple question, as I try to answer this question, I am inevitably humbled by its profoundness.

How do we usually define **similarity** between any different entities: **X** and **Y**?

The concept of similarity

Since we were young, we have learned the concept of similarity by comparing two objects, especially geometric objects, such as triangles.

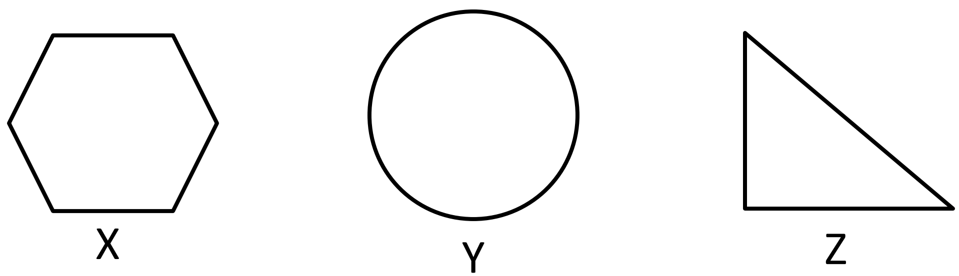


Although X and Y are **not equal**, they are **similar**.

However, what makes them similar, and how do we know they are similar?

The measure of similarity

Now, let us look into **X** and **Y** again. Now, we can see that **X**,**Y** are not similar; neither **X**, **Z**, nor **Z**, **Y**.

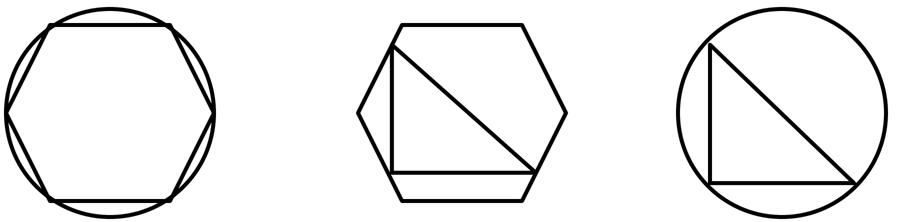


However, it begs for a question as to whether we can still measure the similarity between any two objects, regardless of their shapes?

Actually, we can approach the measure of similarity in two perspectives: **structural** and **informational** perspectives.

The structural similarity

From the structural perspectives, we can partially measure the similarity between any objects by **projection** (overlapping) of their **structural patterns**.



By the projection of their structural patterns onto each other, we can more reliably measure the similarity between **X**, **Y** and **Z**.

The informational similarity

However, informational similarity can measure more generally than structural similarity.

Informational Similarity can be defined such that, if we know **X**, how much we can know about **Y**?

In other words, Informational Similarity measure the **Mutual Information** between any entities.

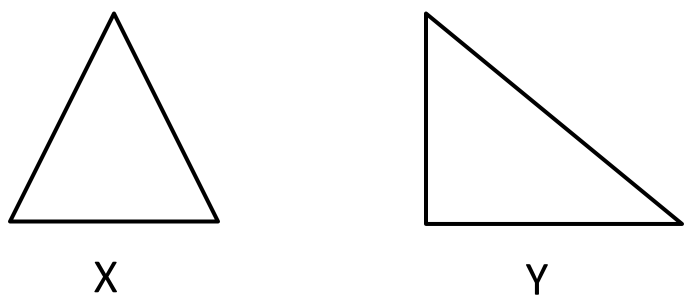
For example, if X is equilateral triangle, and Y is 90% similar to X, then what could Y possibly be?

From this perspective, informational similarity allows us “**Associative Predication**”.

Cross Similarity

Again, we can define similarity between multiple objects as “Cross Similarity”, as to what is similarity between **X** and **Y**?

Cross Similarity can represents the association between any two entities.

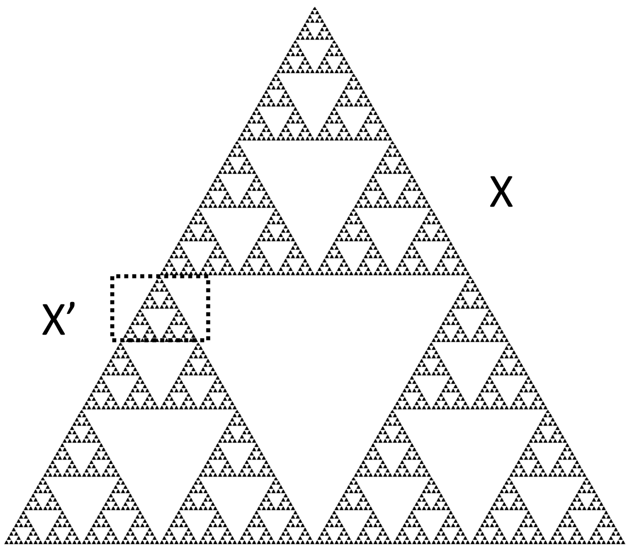


Self Similarity

However, self similarity can be more complicated as it is recursive in nature.

For example, **fractals** are **self-similar** structural objects.

Self similarity can be defined such that if we know partial X', how much we can know about the complete X?



Similarity

Thus, I have a hypothesis that similar entities (objects or experiences) have deducible **similar structural patterns**, and their **mutual information** is **not zero**.

Mutual Information can be defined such that if we know **partial X'**, how much we can know about **Y** or **complete X**: I(X', Y) and I(X', X)

State Space

First, we will look into **Structural Patterns** of Entities (Objects or Experiences).

Mathematically, such **Structural Patterns** can be loosely represented with **State Space**.

State Space can be defined as a “Structured Set”, which is endowed with “Structures”, which can be either linear or non-linear.

Non-Linear Structural State Space can be approximately linearly represented with **Multi-Linear Structural** State Space.

As we know already, such Multi-Linear Structural State Space are **Tensor Space**, which is also the extension of **Vector Space**.

Joint Space

The **Associability** between any two entities X and Y can be represented with **Structural Relationship** which can be represented by **Joint Space** of X and Y.

In other words, the Joint Structural Space of X and Y (**X × Y**) can define the set of all possible structural relationships between X and Y.

However, **Joint Space (A)** of X and Y can be more simply represented by the **Product Space** of the State Spaces of X and Y.

$$f : X \times Y \to A$$

Product Space

Generally speaking, the **Joint Space** between X and Y can be loosely represented by the **Product Space** of X and Y.

However, even in Tensor Space, there are 4 different Product Spaces: **Inner Product Space**, **Outer Product Space**, **Exterior Product Space** and **Interior Product Space**.

All of such product spaces can partially represent Joint Space between X and Y, as they are the subsets of Joint Space of X and Y.

$$X \times Y \rightarrow X \cdot Y$$

$$X \times Y \rightarrow X \otimes Y$$

$$X \times Y \rightarrow X \wedge Y$$

$$X \times Y \rightarrow \iota_X Y$$

Inner and Outer Product Space

Particularly, we will look into Inner and Outer Product Spaces between X and Y. Inner and Outer Product Space are actually the subsets of Joint Space between X and Y.

$$X \times Y \rightarrow X \cdot Y$$

$$X \times Y \rightarrow X \otimes Y$$

Although both Inner Product Space and Outer Product Space can represent the Joint Space between X and Y, the representation itself is different for each product space.

Inner Product Space

Inner Product Space is actually a **Metric Space**, which measures the associative similarity between X and Y in terms of scalar metrics, such as **distance**, **area**, or **volume**.

$$X \times Y \rightarrow X \cdot Y \rightarrow R$$

What is more interesting is Metric Space itself is a **Topological Space**, which can be endowed with “Homeomorphism”.

In addition, all Metric Spaces are Positive or Semi-Positive Space, as negative measurements do not make any sense.

For example, the similarity between two structures (such as geometric shapes) can be loosely represented by their area.

Outer Product Space

Outer Product Space is generally more Interesting than Inner Product Space as it can represent Joint Space between X and, Y in terms of **Mapping**.

$$X \times Y \rightarrow X \otimes Y$$

Actually, **Tensors** are **Multi-Linear Maps** between X and Y, which can represent the Joint Space between X and Y.

The simplest mapping between X and Y can be represented as a **Matrix** (**Rank 2 Tensor**) such that:

$$Ax = y$$

$$A = x \otimes y$$

As we can see, the Matrix can represent the **Structural Association** between X and Y.

Outer Product Space

A **Matrix** is the Map between X and Y, as well as Outer Product Space of X and Y:

$$A = \ x \otimes y$$

$$A = \begin{pmatrix} x_1 y_1 & \cdots & x_n y_1 \\ \vdots & \ddots & \vdots \\ x_1 y_m & \cdots & x_n y_m \end{pmatrix}$$

$$\begin{pmatrix} x_1 y_1 & \cdots & x_n y_1 \\ \vdots & \ddots & \vdots \\ x_1 y_m & \cdots & x_n y_m \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$$

Identity and Similarity

The map between X and Y is **identical** if and only if the Matrix I is the Diagonal Matrix with 1s.

$$I = \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{pmatrix}$$

As we can see, diagonal and non-diagonal entries of the Matrix can represent the similarity measure between X and Y. In fact, even Inner Product Space can be represented by Diagonal Matrix (Non Diagonal Entries are zeros) such that:

$$X \cdot Y = \begin{pmatrix} x_1 y_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & x_n y_n \end{pmatrix}$$

Self Similarity

The entries of a matrix **A** which can measure **self similarity** are known as **Eigen Values**, which is defined as:

$$Ax = \lambda x$$

$$A = x \otimes \lambda x$$

Informational Similarity

As we can see, from the structural perspective, at the simplest level, the Outer Product Space between X and Y can be represented with a Matrix (Rank 2 Tensor), and such matrix can capture the structural relationship between X and Y.

However, from the Informational perspective, how a Matrix could represent such informational similarity.

As I stated before, Informational Similarity can be defined as: with the partial information of **X'**, what is the information of **X** or **Y**?

However, we need to look into **Information** first.

Information

When it comes to information, we will look at it from the perspective of **Entropy**.

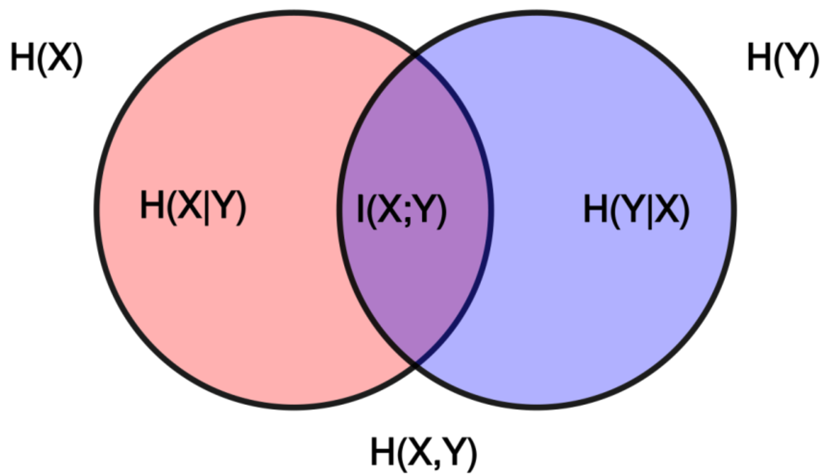
Entropy can be defined as the Average Distribution of Uncertainty in a State Space.

$$H(X) = -\sum P(X)\log\left(P(X)\right)$$

Intuitively, say we have a set of all possible triangles, the information can defined on a set of all possible different distributions of triangles. To put it simply, if we draw a triangle, what is the probability that we might draw an equilateral triangle?

Mutual Information

Now, we will define Mutual Information.



$$I(X,Y) = D_{KL}[P(X,Y) \,||\, P(X) \otimes P(Y)]$$

Mutual Information

Intuitively, we can understand mutual information as follow.

Say, we can make two persons draw “a triangle”, without letting them know about each other. In such case, what is the probability that they could draw the similar triangles?

Say a person X draw a triangle △ABC, and another person Y draw a triangle △ EFG, and there is another person Z, who can only see triangle △ABC drawn by person X. So, how much person Z can know about triangle △ EFG?

How do we define such informational similarity?

Covariance

Formally, we can define a triangle with a state space S, which consists of 3 points in vector space, such that:

$$T: [[x_1, \cdots, x_n], [x_1, \cdots, x_n], [x_1, \cdots, x_n]]$$

For simplification, we will only consider 2 Dimensional Vector Space such that:

$$T: [[x_1, x_2], [x_1, x_2], [x_1, x_2]]$$

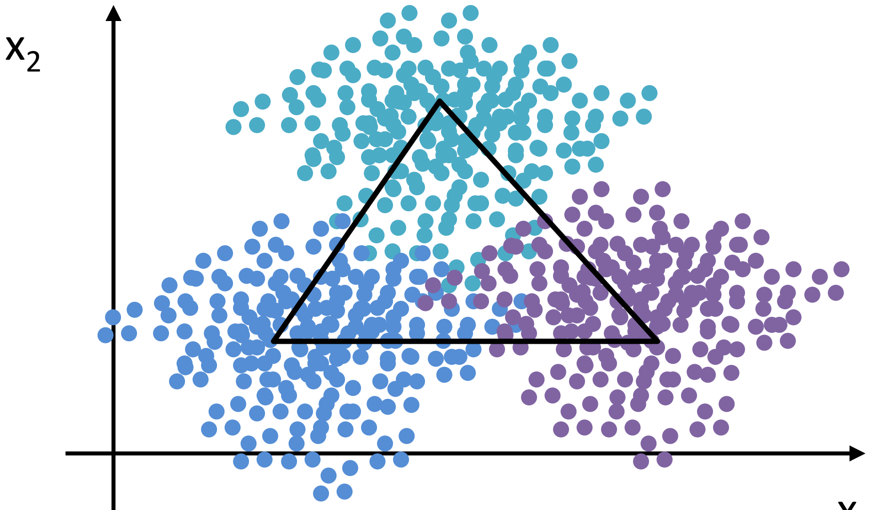
In addition, we will have another constraint such that all three points cannot be collinear.

A random triangle can be represented by a set of 3 random points, which are not collinear, in 2D Space.

Covariance

We can visualize a sample set of all possible triangle data points in 2 Dimensional Space, drawn by a person X.

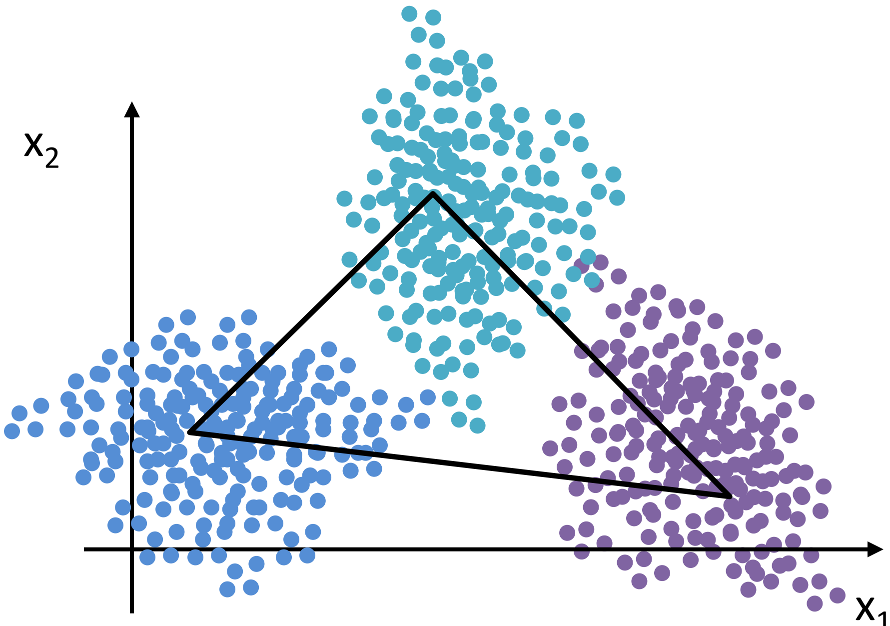
$$T: [[x_1, x_2], [x_1, x_2], [x_1, x_2]]$$



Covariance

We can also visualize a sample set of all possible triangle data points in 2 Dimensional Space, drawn by a person Y.

$$T: [[x_1, x_2], [x_1, x_2], [x_1, x_2]]$$



Covariance

However, to define the Associative Similarity between two sample sets of triangle data points, we need to define the **Covariance**.

As we can see, how similar a triangle drawn by person X is to a triangle drawn by person Y can be represented by the average distance between each points.

However, since there are a lot of possibilities for each point, we will just use the average of the samples, along with their variances deviated from the average.

$$cov(X,Y) = E(X - E(X))E(Y - E(Y))$$

Covariance Matrix

As we can see, **Covariance Matrix** represents the Outer Product Space of Probable Average Samples of X and Y.

$$cov(X,Y) = E(X - E(X))E(Y - E(Y))$$

$$= \begin{pmatrix} E\left(x_1 - E(x_1)\right)E\left(y_1 - E(y_1)\right) & \cdots & E\left(x_n - E(x_n)\right)E\left(y_1 - E(y_1)\right) \\ \vdots & \ddots & \vdots \\ E\left(x_1 - E(x_1)\right)E\left(y_m - E(y_m)\right) & \cdots & E\left(x_n - E(x_n)\right)E\left(y_m - E(y_m)\right) \end{pmatrix}$$

Therefore, Covariance Matrix can measure the **informational similarity**, which is defined by the probability of sample space.

In other words, when we are measuring informational similarity, we are not measuring in the state space; instead we are measuring the average probability in the sample space.

Mutual Information

Loosely speaking, we can define Covariance between X and Y as the mutual information between X and Y:

$$cov(X, Y) \approx I(X, Y)$$

However, such mutual information is not dependent on the Point-wise Mutual Information (PMI). Instead, such mutual information is the measure of the two different distributions.

Actually, it is particularity important simply because Informational Similarity is different from Structural Similarity.

Structural Similarity is defined as to what is the similarity between Triangle A and B, while Informational Similarity is as to what is the similarity between a triangle probably drawn by person A and a triangle probably drawn by person B.

Similarity

As we can see, Similarity is more profound that we expect it to be, as the Generic Definition of Similarity is extremely difficult to define.

At any rate, there are at least two levels in defining similarity:

1. Specifical Similarity
2. Categorical Similarity

The similarity between △ABC and △ EFG is can be defined as Specifical Similarity, in which specific features of particular objects are defined. Likewise, the similarity between Mr. A and Mr. B can be the same.

However, the similarity between Triangle and Rectangle can be defined as Categorical Similarity, in which specific features of particular objects cannot be defined. Another example is the similarity between Chinese and Korean.

Similarity

However, for most cases of Categorical Similarity, we can estimate the similarity by **Mutual Information**.

As we can see, Covariance Matrix is particularly useful in measuring Categorical Similarity, by defining the Average Chinese and Average Korean, and then, we will define the similarity between two averages. Categorical Similarity is the basis of Classification Problem.

Machine Learning

Now, if we ask as to what is Machine Learning with respect to Similarity, we will have to look into the foundational principles of Machine Learning.

Yann et al proposed that there are 4 fundamental aspects in Machine Learning:

1. Output Function
2. Inference Function
3. Loss Function
4. Optimization Function

No matter what types of Neural Architectures we are using, all DNNs are based on these 4 underlying aspects. Interestingly, only Output Function is dependent on Model Architecture, while others are not.

Source: https://atcold.github.io/NYU-DLSP20/en/week07/07-1/

Inference Function

Particularly, *Yann et al* proposed that Inference Function is of primary importance, and thus it is worth looking into what really is the **Inference Function**.

$$Z = argmin_Y F(X, Y, W)$$

To put it simply, Inference Function defines how a Model should evaluate the association between X and Y, given the latent space W.

For example, **Attention Mechanism** is the Inference Function in **Transformer Architecture**. Likewise, **Bi-Directional** (Forward and Backward) Inference is another Inference Function in **Sequence-to-Sequence Architecture**.

Actually, Model Architecture is just to facilitate the Inference Function; in other words, Architecture is the Wrapper of Inference Function.

Energy Function

As Inference Function is of primary importance, and *Yann et al* also suggested that **Energy Function** can be generalized as an **Scalar Inference Function**. Of course, there can be **Vector Inference Function** as well, though a vector itself can be evaluated as a metric scalar.

In its simplest sense, Energy Function can be define as the Measure of Similarity between X and Y such that the more similar X and Y, the less energy it would take to associate between them.

$$I(X,Y) = D_{KL}[P(X,Y) \,||\, P(X) \otimes P(Y)]$$

As we can see, the Mutual Information between X and Y can be defined as a Scalar Energy Function, which is the ratio of **Joint Space** P(X, Y) and **Product Space** P(X) ⊗ P(Y).

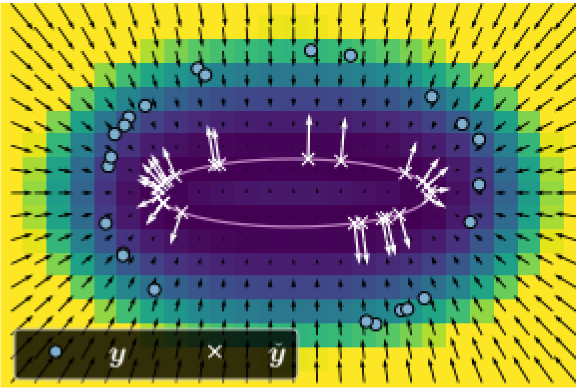
However, "Energy", by definition, is the representation of Entropy. If X and Y are identical, we need zero energy to know Y, when we already know X. If not, we will need more energy.

Manifolds

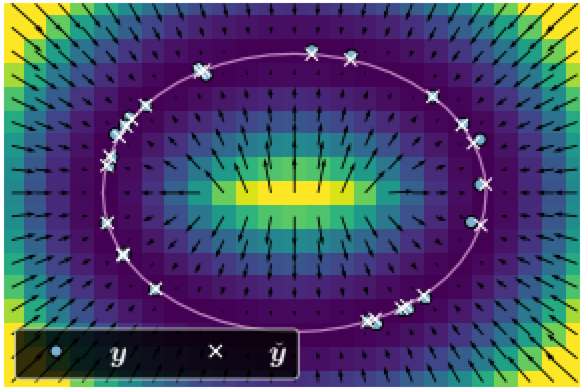
Now, Machine Learning, or more appropriately, Self Entropy Optimization, can be thought of as a process in which Mutual Information is to be maximized (as Energy is minimized) as much as possible between similarities among a set of data points.

Actually, we can think of it as reshaping the Energy Landscape (Manifolds) of Datasets. Obviously, reshaping the Manifolds involves Geometry, which is why Geometric Objects, such as Tensors in Topological Space, have become so important, as Geometry defines the Constraints on such Landscape.

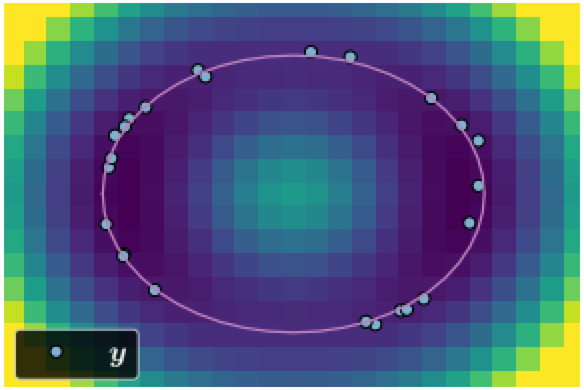
Before Training



During Training



After Training



Source: https://atcold.github.io/NYU-DLSP20/en/week15/15-2/

General Similarity

As we are aware, the core of Machine Learning is Inference Function, and the core of Inference Function is defined by Similarity Measure.

However, General Similarity is extremely hard to define, and in fact, Specifical and Categorical Similarity can be only defined as **Symbolical Similarity**.

It is the major reason why AI works so well with Symbolical Structures, as we can have various sources of symbolical similarity, such as Similar Images or Texts.

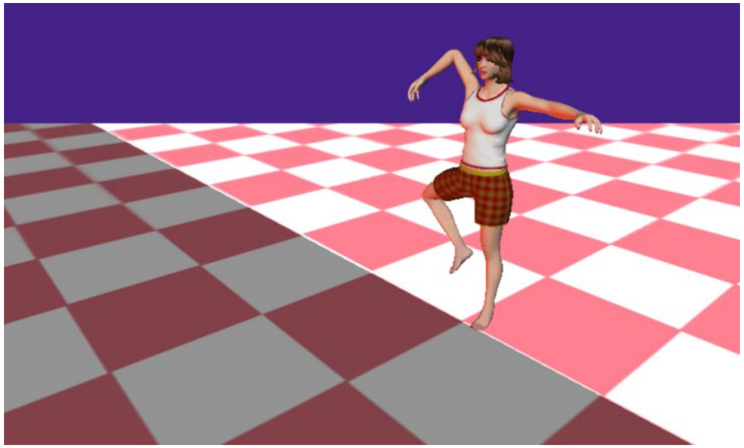
However, **Behavioral Similarity** is extremely hard to define and implement, which is why even the SOTA Systems are struggling with Physical AI, a term coined by *Jensen Huang*.

It might be one of the reasons why *Yann et al* suggested that AI still is not as intelligent as a cat, as it turns out current AI still cannot work well with Physical Behavioral Similarities.

Physical Experience

In the first place, how do we capture the behaviors as data points, such as climbing a tree, swimming in the ocean, eating a pizza, cooking a meal and so on.

Even if we can capture such data points as a set 3D **Point Clouds** or **Postures**, how can we measure the similarity between them, such as what is the similarity between cooking a meal and eating a meal?



https://2kha.github.io/forwardkinematics.html

Towards the Future

What I think the most important aspects of Behaviors are **Actions** and **Evaluations**. And I think the Mappings between Actions and Evaluations can be represented as **Decisions**.

However, the real problem is how do we evaluate all those different actions in real time…

I think a more generic evaluation in Biology could possibly be a **Sentimental Evaluation**. For example: eating a delicious meal would make us as happy as listening to a good music would, simply because the only similar evaluations between them is our sentiments. Other than that, I don’t know how to compare those two actions.

As we can see, measuring similarity in different behaviors is a non trivial task, for which we need a deeper research…